

Sabrina Abram, PhD

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Summary

Research scientist examining the computational and neural mechanisms of adaptive human behavior. My expertise integrates psychology, neuroscience, and machine learning, combining reinforcement learning models with behavioral evidence to identify how the brain solves core challenges of learning. These principles of human intelligence inform the design of human-in-the-loop systems and extend to AI frameworks for advancing interpretability and alignment.

Education

University of California, Berkeley , Postdoctoral Fellow Departments of Psychology and Neuroscience	2022–present
Stanford University , Visiting Scholar Department of Mechanical Engineering	2019–2020
Simon Fraser University , PhD Department of Engineering Science	2015–2021
Cleveland State University , BSc Summa Cum Laude Department of Mechanical Engineering	2011–2015

Experience

10+ years of hypothesis-driven and data-driven science

Cognition and Action Lab, University of California, Berkeley

Advisor: Rich Ivry, Focus: Reinforcement learning in the brain

- Identified systematic biases in human decision making driven by agency in action selection and action execution
- Developed reinforcement learning models to characterize credit assignment mechanisms in neurological patients
- Managed teams across Berkeley, Yale, and Carnegie Mellon, and presented results at 4 international conferences

Biomechatronics Lab, Stanford University

Advisor: Steve Collins, Focus: Human-in-the-loop optimization

- Analyzed longitudinal datasets of human-exoskeleton interaction to reveal explore-exploit strategies in humans
- Identified computational mechanisms and behavioral markers of learning during human-in-the-loop optimization
- Published findings in *Current Biology*, with an invited dispatch article discussing the broader impact of the work

Locomotion Lab, Simon Fraser University

Advisor: Max Donelan, Focus: Energy optimization in human locomotion

- Designed and built a robotic device to test energy optimization as the governing objective of human movement
- Developed computational models of energy optimization to identify learning principles in high-dimensional spaces
- Awarded Canada's top doctoral fellowship for research and leadership; presented at 10 international conferences

Publications

Journal Articles, Book Chapters, and Refereed Conference Proceedings

1. **SJ Abram**, RB Ivry (in press, 2026). Cerebellar contributions to human social cognition. *The Cognitive and Affective Cerebellum. Advances in Neurobiology*.
2. **SJ Abram**, JS Tsay, T Wang, AGE Collins, SD McDougale, RB Ivry (2025). Agency in action selection and action execution produce distinct biases in decision making. *Proceedings of the 6th RLDM Conference*.
3. E Cisneros, K Bol, AGE Collins, **SJ Abram**, RB Ivry, JS Tsay (2025). No evidence for the role of the cerebellum in value-based decision making. *Proceedings of the 6th RLDM Conference*.

4. Y Chen, **SJ Abram**, RB Ivry*, JS Tsay* (2025). Indirect feedback hinders explicit sensorimotor adaptation. *Proceedings of the Royal Society B: Biological Sciences*.
5. **SJ Abram***, JS Tsay*, H Yosef, DS Reisman, HE Kim (2025). The detrimental effect of stroke on motor adaptation. *Neurorehabilitation and Neural Repair*.
6. T Wang, G Avraham, JS Tsay, **SJ Abram**, RB Ivry (2024). Perturbation variability does not influence implicit sensorimotor adaptation. *PLOS Computational Biology*.
7. **SJ Abram**, JS Tsay, T Wang, JJ Li, AGE Collins, SD McDougale, RB Ivry (2024). Differential roles of the cerebellum and basal ganglia in decision making. *Proceedings of the 33rd NCM Conference*.
8. NM van Mastrigt, JS Tsay, T Wang, G Avraham, **SJ Abram**, K van der Kooij, JB Smeets, RB Ivry (2023). Implicit reward-based motor learning. *Experimental Brain Research*.
9. **SJ Abram**, KL Poggensee, N Sánchez, SN Simha, JM Finley, SH Collins, JM Donelan (2022). General variability leads to specific adaptation toward optimal movement policies. *Current Biology*. Featured in Current Biology Dispatch article (<https://doi.org/10.1016/j.cub.2022.03.058>).
10. SN Simha, JD Wong, JC Selinger, **SJ Abram**, JM Donelan (2021). Increasing the gradient of energetic cost does not initiate adaptation in human walking. *Journal of Neurophysiology*.
11. **SJ Abram**, JC Selinger, JM Donelan (2019). Energy optimization is a major objective in the real-time control of step width in human walking. *Journal of Biomechanics*.
12. N Kashiri, A Abate, **SJ Abram**, et al. (2018). An overview on principles for energy efficient robot locomotion. *Frontiers in Robotics and AI*.

Skills

Technical Skills: MATLAB; Python; R; experimental design (lab and online studies); computational modeling (reinforcement learning, maximum likelihood estimation, model simulation, validation, and comparison); statistical methods (mixed-effects models, bootstrapping, permutation testing); time-series analysis and signal processing

Experimental Methods: Measurement of human behavior (eye tracking, motion capture, EMG, metabolic energy expenditure) and brain function (clinical assessment of cognitive and motor function, structural MRI)

Science Communication: 14 presentations at international conferences spanning cognitive science, neuroscience, computer science (AI/ML), and robotics, including invited talks and prize-winning presentations

Leadership: Formally supervised over 10 students (2 Honors) at UC Berkeley and Simon Fraser University

Service

Reviewer: Brain; IEEE Transactions on Neural Systems & Rehabilitation Engineering; Journal of Biomechanics; Journal of Experimental Biology; Neuroscience; Communications Psychology; Experimental Brain Research, NPJ

Mentor: Undergraduate Research Apprentice Program at UC Berkeley; Undergraduate Laboratory Program at UC Berkeley; Women in Science Program at the University of British Columbia; Canadian STEM Science Expo

Committee Member: Society for the Neural Control of Movement

Organizer: Parkinson's Disease Speaker Forum on Brain Function, Movement, and Cognition

Captain: NCAA Division I Women's Soccer; achieved program records in scoring, shutouts, and GPA

Scholarships and Awards

(Selected)

New Investigator Scholarship

2024

Society for the Neural Control of Movement

Michael Smith Foreign Study Supplement

2019–2020

Natural Sciences and Engineering Research Council of Canada

Vanier Canada Graduate Scholarship (\$150,000)

2016–2019

Natural Sciences and Engineering Research Council of Canada

NCAA Division I Women's Soccer Scholarship

2011–2015

Cleveland State University